Homework - Question set 1

Questions on Conditional probability

1) A bin contains 5 defective (that immediately fail when put in use), 10 partially defective (that fail after a couple of hours of use), and 25 acceptable transistors. A transistor is chosen at random from the bin and put into use. If it does not immediately fail, what is the probability it is acceptable?

   Solution: $\frac{5}{7}$

2) The organization that Jones works for is running a father–son dinner for those employees having at least one son. Each of these employees is invited to attend along with his youngest son. If Jones is known to have two children, what is the conditional probability that they are both boys given that he is invited to the dinner? Assume that the sample space S is given by S = (b, b), (b,g), (g, b), (g,g) and all outcomes are equally likely [(b,g) means, for instance, that the younger child is a boy and the older child is a girl]. (**Hint**: Event that Jones is invited to dinner is equivalent event that Jones has at least one son.)

   Solution: $\frac{1}{3}$

3) Ms. Perez figures that there is a 30 percent chance that her company will set up a branch office in Phoenix. If it does, she is 60 percent certain that she will be made manager of this new operation. What is the probability that Perez will be a Phoenix branch office manager?

   Solution: 0.18

Questions on Bayes' formula

4) An insurance company believes that people can be divided into two classes — those that are accident prone and those that are not. Their statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a non-accident-prone person. It is known that 30 percent of the population is accident prone. Suppose that a new policy holder has an accident within a year of purchasing his policy, What is the probability that he is accident prone?

   Solution: 0.461

5) A laboratory blood test is 99 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability a person has the disease given that his test result is positive? (**Note**: In the last sentence, it is 0.5 percent, not probability.)

   Solution: 0.332